

A WPSD-based Feature Extraction Method of EEG Signal for Motor Imagination

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Abstract—In order to improve the performance of the Brain-Computer Interface (BCI) system, it is important to extract the feature vector which reflects non-linearity and non-stationary of Electroencephalogram (EEG) signals. This paper proposed an adaptive feature extraction method for BCI system based on multi-class motor imagery EEG signals. The proposed method extracts a feature vector related with time-frequency-spatial information of multi-class EEG signals from the continuous wavelet transform power spectral density (CWT-PSD) difference weighted by weight matrices which reflect the time-varying of EEG signals, in some channels which reflect well multi-class motor imagery. This method was evaluated on the EEG signals of BCI Competition IV Datasets 2a using SVM as classifier, and the performance results were compared with the results obtained by the Common Spatial Pattern (CSP) and Filter Band CSP (FBCSP) feature extraction method used Naïve Bayesian Parzen Window (NBPW) classifier. The average classification accuracy of the proposed method is 90.22% for the 4-class motor imagery based BCI, and the kappa coefficient of the proposed method is higher 1.38 times and 1.21 times than one of the CSP and FBCSP method, respectively.

Keywords—BCI, EEG, PSD, feature extraction, multi-class motor imagery

I. INTRODUCTION

Brain-Computer Interface (BCI) technology, as one of the directive communication methods between the brain that is the center of all thought activities and the computer, has been actively researched in the world. BCI, as forms a communication channel between a brain and a computer through analysis of the brain activity information generated in the brain, provides an interface that can be used as a control signal by transmitting a person's intention to a computer. BCI is a potential technology that can be applied to various fields, as a system that not only provides the new communication means to persons with disabilities, but also provides convenience to ordinary people [1].

In particular, the BCI based on motor imagery of Electroencephalogram (EEG) signals (MI-BCI) is one of the well-known BCIs. It has been found that motor imagery

directly produces the same EEG signals as when it actually moved in the sensory motor area without moving the part of body. In addition, it has been demonstrated that when it imagines the move or moves a part of body, the Event-Related Desynchronization (ERD) which the power of EEG is relatively decreased because of the asynchronous phenomenon of EEG and the opposite Event-Related Synchronization (ERS) appear in the mu (8~13Hz) band and the beta (13~30Hz) band of the frequency band of EEG signals in the sensory motor cortical area of the brain [2-3].

However, a weak in intensity, the low signal-to-noise ratio, non-linearity, non-stationary of EEG signals still make it difficult to interpret the motor imagery EEG signals. Nevertheless, many efficient feature extraction methods for two or more motor imagery classes (left hand, right hand, foot and tongue) have been developed, overcoming these difficulties. The one of the well-known ones is the Common Spatial Pattern (CSP) method. Based on this method, a number of variant methods have been studied and used in various 2-class or multi-class MI-BCI systems [4].

From the non-linearity and non-stationary of EEG signal, development of an effective feature extraction method with better adaptability and robustness is the key to improve the BCI system performance. This paper proposed an adaptive multi-class feature extraction method by the weighted Power Spectral Density (PSD) difference matrices, denoted as MWDPD (Multi-class feature extraction from the Weighted Difference of PSD). The basic idea of the proposed method is that extracts a feature vector related with time-frequency-spatial information of EEG signals from the PSD difference matrices weighted by weight matrices reflecting the time-varying of EEG signals, in some channels reflecting multi-class motor imagery. Here, the PSD was calculated by Continuous Wavelet Transform (CWT) widely used in the field of intelligent signal processing, as one of time-frequency analysis methods. The method firstly obtains PSD matrices related with time-frequency position according to trails on channels and calculates all possible PSD difference matrices between the channels, and then obtains the weight matrices reflecting time-varying of EEG signals by analysis of PSD difference matrices on some channels which can provide

good classification performance and weights the PSD difference matrices using these weight matrices, finally, extracts the feature vector from the weighted PSD difference matrices [5].

The rest of this paper is organized as follows. Section 2 describes on the EEG signals of BCI Competition IV Datasets 2a used in this study, Section 3 provides the feature extraction method proposed in this paper. Section 4 describes the experimental procedure and gives the experimental results on classification performance of EEG signals in Datasets 2a. Finally, Section 5 concludes this paper with the experimental results.

II. EEG DATA

The BCI Competition IV Datasets 2a is used for this study. This is made up of EEG data from 9 subjects. The Prompt-Based BCI paradigm consists of four different motor imagery tasks, in other words, the imagination of movement of the left hand (Class 1), right hand (Class 2), both feet (Class 3), and tongue (Class 4). It was recorded in two sessions on different days for each subject, and then each session is comprised of 6 runs separated by short breaks. One run consisted of 48 trials (12 of each of four possible classes), resulting in a total of 288 trials per a session. The subject was sitting in a comfortable armchair in front of the computer screen. At the beginning of the trial ($t = 0s$), a fixed cross appears on the black screen. After two seconds ($t = 2s$), a cue in the form of an arrow pointing either to the left, right, bottom or top (corresponding to one of the four classes left hand, right hand, foot or tongue) appeared and remained on the screen for 1.25 seconds. This contributes the subject to perform the desired motion imagery task. No feedback was provided. The subject is asked to perform a motion imaging task until the fixed crossover disappears from the screen at $t = 6s$. A short break, followed by the screen again black. The EEG is recorded using twenty-two Ag/AgCl electrodes with an inter-electrode distance of 3.5cm. All signals were mono-polar recordings. The signals were sampled at 250Hz and band pass-filtered between 0.5Hz and 100Hz. An additional 50Hz notch filter was enabled to suppress line noise [6].

III. METHOD

A. Power Spectral Density by Continuous Wavelet Transform

Continuous Wavelet Transform (CWT) can provide a very redundant and detailed signal description in time and frequency domain. Given a time series signal $x(t)$, its CWT coefficient $W_\psi(\tau, s)$ is defined as follows [7].

$$W_\psi(\tau, s) = \int_{-\infty}^{\infty} x(t) \frac{1}{\sqrt{|s|}} \psi^*\left(\frac{t-\tau}{s}\right) dt, \quad (1)$$

where, $\psi^*(t)$ is a complex conjugate function of a wavelet basis function $\psi(t)$, τ and s relate with the positions in the time and frequency, respectively. As seen in Equation (1), WT provides simultaneously the information on time and frequency scale, by mapping the input signals into the function of τ and s .

The PSD of CWT $W_\psi(\tau, s)$ is defined as follows.

$$P(\tau, s) = |W_\psi(\tau, s)|^2 \quad (2)$$

From Equation (2), the power distribution of signals according to time and frequency can be known.

B. Davies-Bouldin Index

To evaluate the degree of class separation, Davies-Bouldin Index is calculated by measuring the overlap degree of two classes in any dimensional space. When f is one element of a vector with N samples, DB-Index is calculated through following equations [8].

$$m_i = \frac{1}{N_i} \sum_{i=1}^{N_i} f, \quad (3)$$

$$s_i = \frac{1}{N_i} \sum_{i=1}^{N_i} |f - m_i|, \quad (4)$$

$$R_{ij} = \frac{s_i + s_j}{|m_i - m_j|}, \quad (i \neq j), \quad (5)$$

where, N_i is the number of samples belonging to class C_i among the N samples of $f, i=1, 2, \dots, N_c, j=1, 2, \dots, N_c, N_c$ is the number of classes. R_{ij} represents the overlap degree of samples belonging to class C_i and C_j by using the centroid and the spreads of each in two classes. The smaller the index value, the less overlapping the samples in two classes, and the larger the index value, the more overlapping the samples in two classes. The matrix R with R_{ij} as the elements in all possible class combinations when $N_c=4$ is constructed as follows.

$$R = \begin{bmatrix} - & R_{12} & R_{13} & R_{14} \\ R_{21} & - & R_{23} & R_{24} \\ R_{31} & R_{32} & - & R_{34} \\ R_{41} & R_{42} & R_{43} & - \end{bmatrix} \quad (6)$$

C. Channel Selection by DB-Index

The ERD / ERS phenomena are the physiological phenomena to accurately classify motor imagery in EEG signals. However, ERD and ERS phenomena for any motor imagery appear in some brain area, for instance, when motor imagery of right hand, ERD appears in the left side sensory motor area of cerebral cortex, while ERS appears in the right side one [2-3]. Therefore, each channel has a different contribution to the classification performance of some motor imagery. In order to decrease the computational complexity while increasing the classification performance of MI-BCI system, it is necessary to select the channels which reflect the ERD / ERS phenomena of motor imageries well.

In this study, from the characteristics of ERD / ERS phenomena, it was used the difference of PSDs between two channels instead of PSD in each channel. These the difference of PSD between all possible two channels is expressed as follows.

$$dP_{i,j}(\tau, s) = P_{m,j}(\tau, s) - P_{n,j}(\tau, s), \quad (m \neq n), \quad (7)$$

where, $m=1, 2, \dots, N_h$, $n=1, 2, \dots, N_h$, N_h is the number of EEG channels, $j=1, 2, \dots, N_i$, N_i is the number of trials in a EEG channel, $i=1, 2, \dots, N_d$, N_d is equal to $C(N_h, 2)$ as the number of all possible combination of two channels.

Now, it can be obtained DB-Index between a class and rest classes according to trials for the every element of $dP_{ij}(\tau, s)$. This result is a matrix of the same dimensions as the matrix $dP_{ij}(\tau, s)$. Let $R_{i,k}(\tau, s)$ be this result. $R_{i,k}(\tau, s)$ represents the DB-Index value between the k th class and the rest classes according to trials for the power of EEG signal at the time τ position and the frequency s position in the PSD difference of the i th channel combination. Provided that the number of classes is N_c , k is 1 to N_c . From the DB-Index definition in Equation (5), it can be seen that the channel combination corresponding to a matrix containing smaller element values among matrix set $\{R_{i,k}(\tau, s)\}$ ($i=1, \dots, N_d$, $k=1, \dots, N_c$) contributes to a good classification performance of MI-BCI system. Thus, a matrix set of matrices containing the smallest elements according to the channel combination for each class can be obtained. Let be $\{R_1(\tau, s), R_2(\tau, s), \dots, R_{N_c}(\tau, s)\}$ this matrix set. Here, $R_k(\tau, s)$ ($k=1, 2, \dots, N_c$) is a matrix selected in a matrix set $\{R_{i,k}(\tau, s)\}$. Additionally, the channel combinations corresponding to selected matrices can also be obtained.

D. Weight Matrix Calculation by DB-Index

The feature vector can be extracted from the PSD differences in the channel combinations selected for a good classification of each class. However, it is necessary again to consider DB-Index matrices selected above before feature extraction. These index matrices indicate the degree which the distribution of the power difference in two channels overlaps between a certain class and the rest classes in the entire trials. The smaller index values represent the less overlapping. From these index matrices, the following weight matrices can be considered.

$$W_k(\tau, s) = \frac{1}{R_k(\tau, s)}, \quad (k=1, 2, \dots, N_c), \quad (8)$$

where, k is a class index, $R_k(\tau, s)$ is a DB-Index matrix selected for k th class. The weight matrix has a very profound meaning, which reflects the variation of the distribution of PSD differences in corresponding to channel combination according to trials and classes. In other words, the weight matrix reflects how well PSD differences are classifiable the multi-class motor imagery in the total trials. So, the feature vector can be extracted from the weighted PSD differences obtained by multiplying the corresponding elements of the PSD difference matrix and the weight matrix as follows.

$$wP_k(\tau, s) = W_k(\tau, s) \cdot P_k(\tau, s) \quad (9)$$

where, $W_k(\tau, s)$ and $P_k(\tau, s)$ are the weight matrix and the PSD difference matrix selected for k th class motor imagery, k is the class index. The procedure that extracts the feature vector from the weighted PSD difference is described in following subsection.

E. Feature Extraction

From the above processes, the weighted PSD differences

$wP_k(\tau, s)$ ($k=1, 2, \dots, N_c$) were obtained. In this our study, the features reflecting frequency and time information were extracted from the weighted PSD differences by k -means clustering method. The k -means method firstly separates the weight matrix as input into two groups and results separation index corresponding to time scale, and then separates the transpose of weight matrix as input into two groups and results separation index corresponding to frequency scale. From the meaning of the weight matrix, only the group with the largest center value among the two groups resulted by k -means was used for feature extraction for this study. Let IT_k and IF_k denote the group index with the largest center value among each two groups related with time and frequency scale for the k th weight matrix. Then, the feature vector is configured as follows.

$$fT_k = \sum_{\tau=IT_k} \sum_s wP_k(\tau, s), \quad (10)$$

$$fF_k = \sum_{s=IF_k} \sum_{\tau} wP_k(\tau, s), \quad (11)$$

$$F = [fF_1, fT_1, \dots, fF_{N_c}, fT_{N_c}]^T, \quad (12)$$

where, fT_k is a feature value obtained by summing the PSD difference according to IT_k time positions and entire frequency positions in selected k th channel combination, likewise, fF_k is a feature value related with IF_k frequency positions and entire frequency positions. A feature vector is configured as the feature values related with time and frequency obtained in each channel combination. The proposed feature extraction method is shown in Figure 1.

IV. EXPERIMENT AND RESULT

The performance experiment of the multi-class feature extraction method proposed in this paper, MWDPSD, was evaluated on the BCI Competition IV Datasets 2a described in Section 2. The performance results were evaluated by kappa coefficient and classification accuracy. The kappa coefficient could be calculated by Equation (13).

$$kappa = \frac{aN_c - 1}{N_c - 1}, \quad (13)$$

where, a is the classification accuracy and N_c is the number of classes.

Many classification algorithms for MI-BCI have been researched and used. Among them, Support Vector Machine (SVM) classifier shows good results not only in 2-class motor imagery classification but also in multi-class problem. This is because it has a good generalization capability, the robustness with respect to the dimensionality of feature vectors. Therefore, the SVM was used as classifier for our experimental research.

Performance results of our method were compared with the results obtained by extracting the multi-class feature as a one-versus-rest mode of CSP and FBCSP method and using the Naïve Bayesian Parzen Window (NBPW) method as a classifier. Table 1 shows the performance result as the classification accuracy and the kappa coefficient, and Figure 2 shows the performance comparison chart of three feature extraction methods.

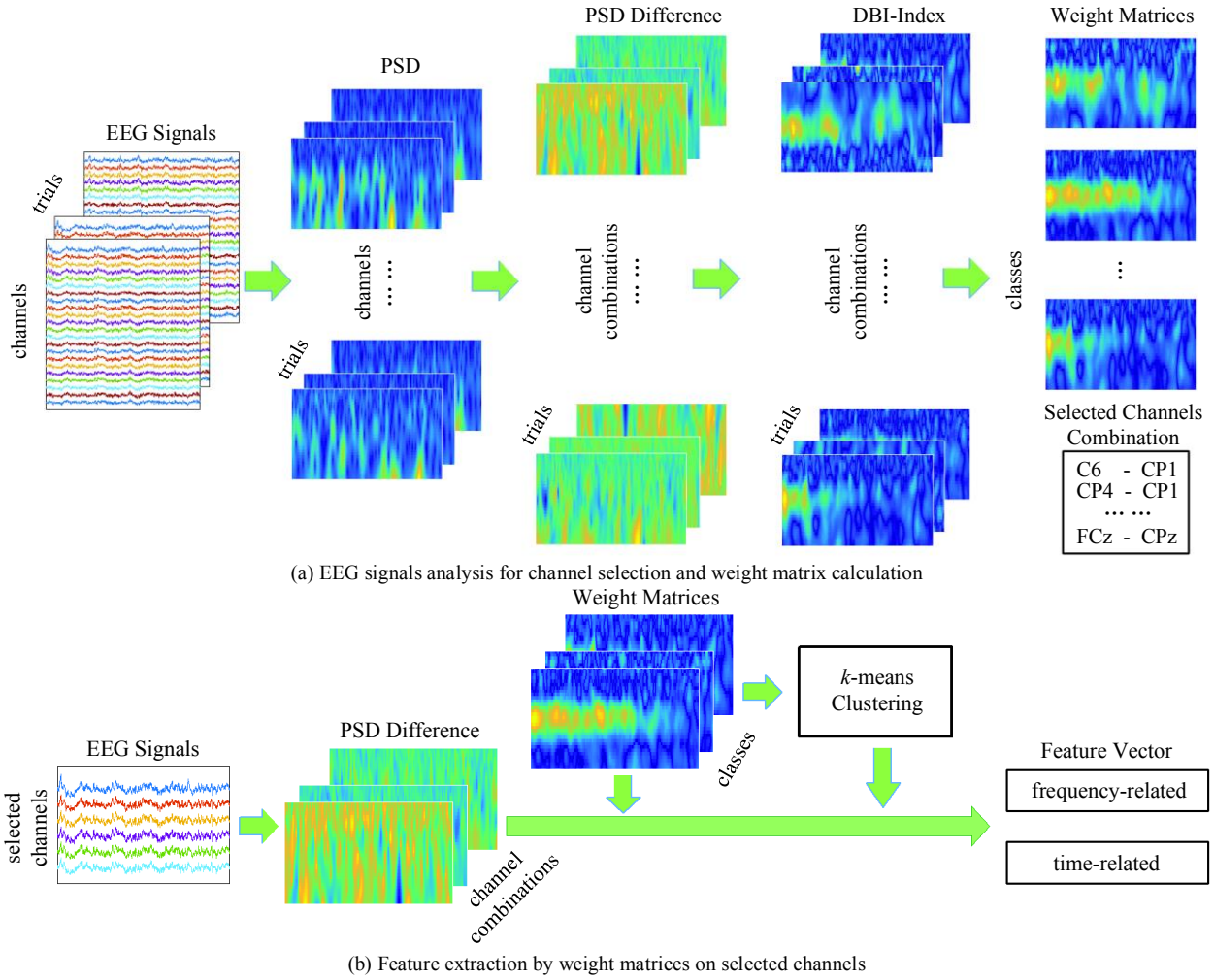


Figure 1. The overall processing scheme for feature extraction of multi-class EEG signals

TABLE I. THE PERFORMANCE RESULTS OF THE PROPOSED MWDPSP, CSP AND FBCSP METHODS

Subjects	Accuracy (%)			kappa		
	CSP	FBCSP	MWDPSP	CSP	FBCSP	MWDPSP
A01	73.30	81.93	93.63	0.644	0.759	0.915
A02	56.73	60.63	92.59	0.423	0.475	0.901
A03	84.78	87.55	88.93	0.797	0.834	0.852
A04	52.38	61.30	90.99	0.365	0.484	0.880
A05	41.13	70.08	86.99	0.215	0.601	0.827
A06	46.00	51.03	89.41	0.280	0.347	0.859
A07	71.95	89.65	90.22	0.626	0.862	0.870
A08	83.05	85.53	85.76	0.774	0.807	0.810
A09	78.93	84.10	93.50	0.719	0.788	0.913
Avg	65.36	74.64	90.22	0.538	0.663	0.870

As seen in figure, the classification accuracy of our method is 1.38 times and 1.21 times higher than one of the CSP and FBCSP method, respectively. Also, kappa coefficient of our method is 1.62 times and 1.31 times higher than one of them.

V. CONCLUSION

This paper proposed an adaptive feature extraction method for multi-class MI-BCI system. In this paper, PSD was calculated by using CWT, one of time-frequency analysis methods for analyzing non-linear, non-stationary signal. Several channels with high contribution for multi-class motor imagery classification were selected by using DB-Index, and the PSD differences were calculated on these channels to reflect spatial information of EEG signals. The weight matrices corresponding to PSD differences, which reflect the change in the power value of the EEG signals according to time and trial, were also calculated by the DB-Index. From these, the feature vector that reflects time and frequency information was extracted from the weighted PSD differences. The feature extraction process was shown that the proposed method reflects not only the time-frequency-spatial information of the EEG signals, but also the time-varying.

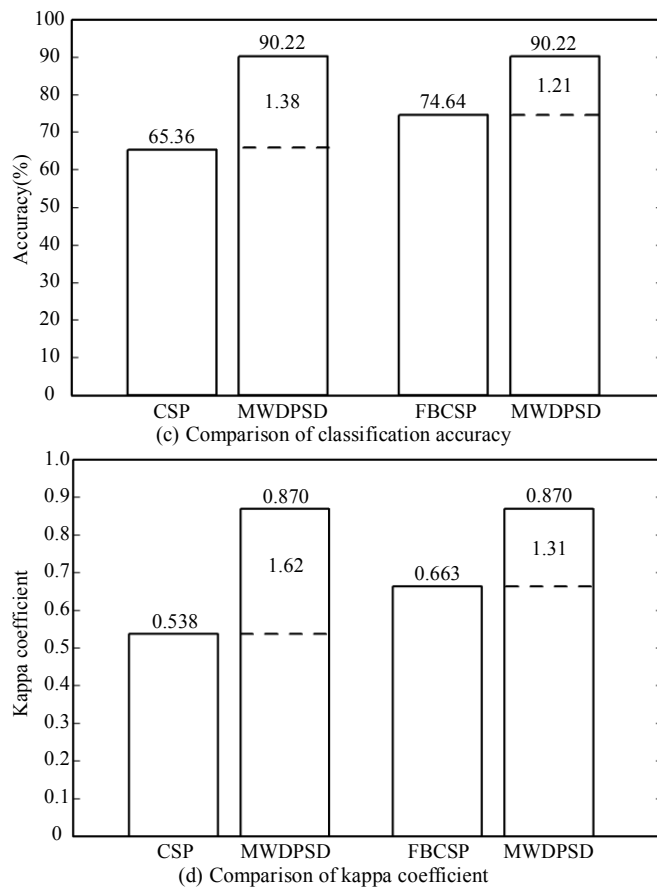


Figure 2. The performance comparison of the proposed MWDPSD, CSP and FBCSP methods

The experimental results show that the proposed method results a good classification performance with the average classification accuracy of 90% for 4-class motor imagery classification. Compared with CSP and FBCSP feature extraction methods, this method has 1.62 times and 1.31 times larger kappa coefficients, respectively. To sum up, this method

has some potential for extracting feature of EEG signals in multi-class MI-BCI system.

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